

CAAM/STAT31470: Homework #3

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: On this homework, we investigate the properties of elementary functions of complex variables. In particular, we look at their properties as multivalued functions. We also look at some first examples of integration along paths.

Homework is due: **Wednesday April 16, 11:59pm.**

Problem 3.1 (8 points)

- (a) Use the definition of z^c to show that $(-1 + \sqrt{3}i)^{3/2}$ is $\pm 2\sqrt{2}$. Show that you get the same answer by writing
- (i) $[(-1 + \sqrt{3}i)^{1/2}]^3$ and first finding the square roots; and
 - (ii) $[(-1 + \sqrt{3}i)^3]^{1/2}$ and first finding the cube.
- (b) Let $c = a + bi$ be a fixed complex number where $c \neq 0, \pm 1, \pm 2, \dots$. Show that i^c is multivalued. Also what restriction is needed on c so that $|i^c|$ is constant for all c .
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Problem 3.2 (6 points)

Verify the formula: $\sin^2(z) + \cos^2(z) = 1$; in the following two ways.

- (a) From the definition. First compute $e^{iz_1}e^{iz_2}$ and $e^{-iz_1}e^{-iz_2}$ in terms of the sine and cosine of z_1 and z_2 . Using those expressions verify that

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin(z_1)\sin(z_2).$$

Finally set $z_1 = z$ and $z_2 = -z$.

- (b) Define the function $f(z) = \sin^2(z) + \cos^2(z) - 1$ and show that it is analytic for all $z \in \mathbb{C}$. By looking at the values of the function on the real line, conclude that $f(z)$ is a constant function.
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Problem 3.3 (8 points)

- (a) Find all solutions z for the equation:

$$\sin(z) = \cosh(4);$$

by equating real and imaginary parts.

- (b) Find formulas for the inverse trigonometric function $\cos^{-1}(z)$. Hint: The goal is to solve $z = \cos(w)$ for w , and you can use the definition of sine to find a quadratic equation in e^{iw} . Using the Cauchy-Riemann equations, show that $\sin^{-1}$ is an analytic function on a domain containing the imaginary axis.
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Problem 3.4 (8 points)

For each of the following arcs:

- (i) $z(t) = e^{it}$ for $t \in [0, \pi/2]$
- (ii) $z(t) = 1 + it$ for $t \in [0, 1]$ and $2 - t + i$ for $t \in [1, 2]$
- (iii) $z(t) = e^{-it}$ for $t \in [0, 3\pi/2]$

plot the arc and evaluate the integral

(1)
$$\int_{t_1}^{t_2} \sqrt{z(t)} dt$$

where the integration is over the natural domain of the arc.

End of homework